1.

Consider the following questions:

Find two non-empty sets, A, B, such that $A \in B$ but $A \nsubseteq B$

Let $A = \{1\}, B = \{\{1\}\}$

$A \in B$, $A \nsubseteq B$

Find two non-empty sets, A, B, such that $A \notin B$ but $A \subseteq B$

Let $A = \{-1, -2\}, B = \{-1, -2, \{-1\}\}$

2.

Let $\{A_i\}_{i \in I}$ be a family of sets indexed by I in a universe $\mathcal{U}$. Prove that $\overline{\bigcup_{i \in I} A_i} = \bigcap_{i \in I} \overline{A_i}$

- $\overline{\bigcup_{i \in I} A_i} = \overline{\{A_0 \cup ... \cup A_i\}}$
- $\neg\{A_0 \vee ... \vee A_i\}$
- $\{\neg A_0 \wedge ... \wedge \neg A_i\}$
- $\left\{\overline{A_0} \cap ... \cap \overline{A_i}\right\}$
- $\bigcap_{i \in I} \overline{A_i}$

3.

Write the power set of $S = \{a, b, c\}$

$\mathcal{P}(S) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}\{a, c\}, \{b, c\}, \{a, b, c\}\}$

4.

Let $A = \{1, 2\}, B = \{2, 3\}$

a. Compute $|A|, |B|, |A \cup B|, |A \cap B|$.

$|A| = 2$

$|B| = 2$

$|A \cup B| = 3$

$|A \cap B| = 1$

b. Write out $\mathcal{P}(A \cup B)$ and verify that $|\mathcal{P}(A \cup B)| = 2^{|A \cup B|}$

$\mathcal{P}(A \cup B) = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$

$|\mathcal{P}(A \cup B)| = 8$

$2^{|A \cup B|} = 2^3 = 8$

$|\mathcal{P}(A \cup B)| \equiv 2^{|A \cup B|}$

5.

For the sets $A = \{a\}$ and $B = \{a, b\}$, write down the following:

a. $\mathcal{P}(A) = \{\varnothing, \{a\}\}$

b. $\mathcal{P}(B) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}$

c. $\mathcal{P}(B \setminus A) = \{\varnothing, \{b\}\}$

d. $\mathcal{P}(B) \setminus \mathcal{P}(A) = \{\{b\}, \{a, b\}\}$

e. $A \times \mathcal{P}(B) = \{(a, \varnothing), (a, \{a\}), (a, \{b\}), (a, \{a, b\})\}$

f. $\mathcal{P}(A \times B) = \{\varnothing, \{(a, a)\}, \{(a, b)\}, \{(a, a), (b, b)\}\}$

6.

For the same sets above, which of the following are true?

a. $A \in \mathcal{P}(B) : \top$

b. $A \subseteq \mathcal{P}(B) : \bot$

c. $\mathcal{P}(A) \in \mathcal{P}(B) : \top$

d. $\mathcal{P}(A) \subseteq \mathcal{P}(B) : \top$

7. Let A, B be arbitrary sets (which could be empty). If $A \subseteq B$, which of the four statements in the above question hold in general? Prove the ones that do, and find counterexamples for the others.

a.

- $\mathcal{P}(B)$ is defined as the set of all possible subsets of B
- As $A \subseteq B$, A is part of the possible subsets of B
- Therefore, $A \in \mathcal{P}(B)$

b.

- Let $A = \{a\}, B = \{a, b\}$
- $\mathcal{P}(B) = \{\varnothing, \{a\}, \{b\}, \{a, b\}$
- $A \nsubseteq \mathcal{P}(B)$

c.

- Let $A = \{1\}, B = \{1\}$
- $\mathcal{P}(A) = \{\varnothing, \{1\}\}, \mathcal{P}(B) = \{\varnothing, \{1\}\}$
- $\mathcal{P}(A) \notin \mathcal{P}(B)$

d.

- Let $A = \{1\}, B = \{1\}$
- $\mathcal{P}(A) = \{\varnothing, \{1\}\}, \mathcal{P}(B) = \{\varnothing, \{1\}\}$
- $\mathcal{P}(A) \subseteq \mathcal{P}(B)$

8. Answer the following questions on Cartesian product:

a. Is the Cartesian product commutative? What about associative?

Let $A = \{1, 2\}, B = \{3, 4\}, C = \{5\}$

$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$

$B \times A = \{(3, 1), (3, 2), (4, 1), (4, 2)\}$

$B \times C = \{(3, 5), (4, 5)\}$

$(A \times B) \times C = \{((1, 3), 5), ((1, 4), 5), ((2, 3), 5), ((2, 4), 5)\}$

$A \times (B \times C) = \{(1, (3, 5)), (1, (4, 5)), (2, (3, 5)), (2, (4, 5))\}$

- The Cartesian product is not commutative, as the product produces ordered pairs, which are changed depending on the order of operands.
- The Cartesian product is not associative, as ordered pairs differ when changing the grouping of terms.

b. Show that for sets A, B we have $(A \times B = \varnothing) \leftrightarrow ((A = \varnothing) \vee (B = \varnothing))$

Case 1: $(A \times B = \varnothing) \Rightarrow ((A = \varnothing) \vee (B = \varnothing))$

- The Cartesian product of A, B is the set of all ordered pairs between elements of A, B
- There must be at least 2 elements in an ordered pair
- Therefore, if $A \times B = \varnothing$, then $A = \varnothing \vee B = \varnothing$, as only one of the sets need to be empty, as it will prevent ordered pairs from forming.

Case 2: $((A = \varnothing) \vee (B = \varnothing)) \Rightarrow (A \times B = \varnothing)$

- The $\varnothing$ consists of no elements. If at least one of A, B consist of no elements, this means that no pairs can be formed, thus $A \times B = \varnothing$
- **Therefore, $(A \times B = \varnothing) \leftrightarrow ((A = \varnothing) \vee (B = \varnothing))$**

c. What can be said, for any sets A, B, about $(A \times B) \cap (B \times A)$

- An ordered pair (x, y) only exists in $(A \times B) \cap (B \times A)$ when $X \in A, y \in B \wedge x \in B, y \in A$
- Therefore, $(A \times B) \cap (B \times A) = (A \cap B) \times (B \cap A)$

d. Can you find example sets A, B so that both $A \subseteq B$ and $A \subseteq \mathcal{P}(B)$

- Let $A = \varnothing, B = \{1\}$
- $\mathcal{P}(B) = \{\varnothing, \{1\}\}$
- $A \subseteq B$ and $A \subseteq \mathcal{P}(B)$